

BIOPHYSICAL NETWORK

The `Neuronal_Network` function sets up a network of neurons. It uses a Hodgkin-Huxley-like model (`eqs_NN`) for individual neurons, incorporating parameters for sodium, potassium, and AHP currents, as well as a noise term. It then defines the synapses (`eqs_Syn`), which can follow either a Tsodyks model (for synaptic depression/facilitation) or a kinetic model. The function handles both random and distance-based connectivity and can optionally add conduction delays.

The `Astrocyte_Group` function creates a network of astrocytes, which are modeled using a detailed set of equations (`eqs_A`) that describe calcium and IP3 dynamics. Astrocytes are connected by gap junctions, allowing for the diffusion of IP3.

The `Gliotransmission` function models the release of gliotransmitters by astrocytes. The release is triggered when the intracellular calcium concentration exceeds a certain threshold.

The final two functions, `Synapse_to_astro` and `Astro_to_Syn`, define the connections that link the neuronal and astrocytic networks. `Synapse_to_astro` models the effect of synaptic activity (neurotransmitter release) on astrocytes, and `Astro_to_Syn` models the effect of gliotransmitter release on synapses, completing the bidirectional communication pathway.

This is a complete script for simulating a complex neuro-astrocytic network.

Key Functions and Their Roles

Neuronal_Network

This function is the main component for setting up the neuronal network.

- **Neuron Model (`eqs_NN`):** It uses a variant of the Hodgkin-Huxley model, including currents for sodium (`g_na`), delayed-rectifier potassium (`g_kd`), and a slow after-hyperpolarization (`g_AHP`) current, as well as a leaky current (`gl`). A noise term is added to simulate intrinsic neuronal variability.
- **Synapse Model (`eqs_Syn`):** It provides two options for synaptic currents:
 - **Nina's model:** A phenomenological model with separate dynamics for AMPA and NMDA receptor activation, incorporating short-term synaptic plasticity (`x_d`, `s_nmda`, `x_nmda`).
 - **TM-coupled model:** A more detailed kinetic model that couples the postsynaptic response to the presynaptic neurotransmitter concentration (`Y_S`).
- **Connectivity:** It supports both **Random** connectivity, where a connection is formed with a given probability, and **Distance-based** connectivity, where the probability of connection decreases with the distance between neurons. The function uses the `Distance_based_connections` and `unfold_ADJ` functions defined in the previous part of the script to create and interpret the adjacency matrix.

Astrocytic Network Functions

Astrocyte_Group

This function models the astrocytic network and their internal dynamics.

- **Astrocyte Model (eqs_A):** The model describes the dynamics of intracellular IP3 (I), calcium (C), and activated receptors (Gamma_A). It includes IP3 production from presynaptic activity (Y_extra), IP3 diffusion between astrocytes via gap junctions (I_coupling_tot), and calcium-induced calcium release from the ER.
- **Connectivity (Gap_Eq):** It establishes **Gap-junctions** between astrocytes, allowing for the diffusion of IP3. This is based on a distance-dependent connectivity scheme.
- **External Stimulation:** When Simulated_network is 'Astrocytic', it sets up an external stimulus via a Poisson process to trigger astrocyte activity.

Gliotransmission

This function is specifically for modeling gliotransmitter release.

- **Release Mechanism:** It's an event-based model where gliotransmitter release (G_A) is triggered when the astrocytic calcium concentration (C) crosses a defined threshold (C_Theta). This effectively translates calcium oscillations into discrete gliotransmitter release events.

Neuro-Astrocyte Communication

This section of the code handles the crucial communication between the two networks.

Synapse_to_astro

This function connects synapses to astrocytes. It models the effect of presynaptic activity on astrocytes by linking the **extracellular neurotransmitter concentration** (Y_S_pre) at the synapse to the astrocyte's Y_extra_post variable. This allows synaptic activity to drive IP3 production in the connected astrocyte.

Astro_to_Syn

This function closes the loop by connecting the astrocytes back to the synapses. It links the **released gliotransmitter** (G_A_pre) from the Gliotransmission group to the G_A_post variable in the synapse. In the full model, this gliotransmitter concentration would then modulate the synaptic release probability (U_0), completing the tripartite synapse feedback loop.

Astrocyte Parameters (get_Astroparam)

These parameters define the dynamic behavior of astrocytes, their calcium signaling, and their interaction with the neuronal network. The dynamics are governed by a complex interplay of ion channels, pumps, and messenger molecules.

- **f_in:** Input frequency from neurons (or a stand-alone simulation). It drives the production of IP3.
- **f_c:** Intrinsic oscillation frequency of the astrocyte.
- **t_off:** End time for astrocyte stimulation.

- **d_1, d_2, d_3, d_5: IP₃ receptor affinities.** These parameters define how IP₃ and calcium (Ca²⁺) bind to the IP₃ receptors on the Endoplasmic Reticulum (ER) membrane. This binding is a crucial step for triggering Ca²⁺ release from the ER.
- **O_2: Ca²⁺ inactivation rate.** This parameter determines how quickly Ca²⁺ itself can inactivate the IP₃ receptor, providing a negative feedback loop.
- **C_osc:** Threshold for detecting calcium oscillations.
- **C_T: Total Ca²⁺ concentration** inside the ER. This acts as the maximum pool of Ca²⁺ that can be released.
- **rho_A:** Ratio of the ER volume to the cytoplasm volume. It scales the Ca²⁺ concentration changes.
- **Omega_C:** Maximal rate of Ca²⁺ release from the ER.
- **Omega_L:** Maximal rate of Ca²⁺ leakage from the ER.
- **O_P:** Maximal rate of Ca²⁺ uptake by SERCA pumps, which pump Ca²⁺ back into the ER. This is a key parameter for clearing Ca²⁺ from the cytoplasm.
- **K_P:** Ca²⁺ affinity of the SERCA pumps. A lower value means the pump is more efficient at lower Ca²⁺ concentrations. It distinguishes **AM** (Amplitude Modulation) and **FM** (Frequency Modulation) oscillation modes. In AM mode, the amplitude of Ca²⁺ oscillations varies, while in FM mode, the frequency of oscillations changes with input intensity.
- **O_beta:** Maximal rate of IP₃ production in response to neuronal activity. It represents the link from neurons to astrocytes.
- **O_delta:** Maximal rate of Ca²⁺-dependent IP₃ production. This creates a positive feedback loop where high Ca²⁺ can generate more IP₃.
- **K_delta:** Ca²⁺ affinity of PLC δ .
- **kappa_delta:** Inhibiting IP₃ affinity of PLC δ .
- **O_3K:** Maximal rate of IP₃ degradation.
- **K_D:** Ca²⁺ affinity of IP₃-3K.
- **K_3K:** IP₃ affinity of IP₃-3K.
- **F: Gap junction permeability.** This parameter controls the rate of IP₃ diffusion between coupled astrocytes, enabling intercellular calcium waves.
- **I_Theta:** Threshold IP₃ gradient for diffusion.
- **omega_I:** Scaling factor of diffusion.
- **O_N: Agonist binding rate.** It represents how quickly astrocytic receptors bind to neurotransmitters.
- **Omega_N:** Inactivation rate of GPCR signaling.
- **K_KC:** Ca²⁺ affinity of PKC.
- **zeta:** Maximal reduction of receptor affinity by PKC.
- **n:** Cooperativity of agonist binding reaction.

- **C_Theta: Ca²⁺ threshold for gliotransmitter release.** When cytoplasmic Ca²⁺ exceeds this value, an astrocytic 'spike' occurs, and gliotransmitter is released.
- **Omega_A:** Gliotransmitter recycling rate.
- **U_A:** Gliotransmitter release probability per event.
- **G_T:** Total vesicular gliotransmitter concentration.
- **rho_e:** Ratio of astrocytic vesicle volume to extracellular space (ECS) volume.
- **Omega_e:** Gliotransmitter clearance rate from the ECS.
- **spill_over:** A factor for how much neurotransmitter from a synapse spills over into the extracellular space to reach an astrocyte.
- **conn_dist:** Maximum distance for astrocytic gap junction connections.
- **c_min, c_max:** Coordinates for placing astrocytes in a 2D space.

Neuron Parameters (get_Neuronparam)

These parameters control the excitability, firing patterns, and other biophysical properties of the neurons.

- **area:** Membrane surface area, which scales the conductances and capacitance.
- **Cm: Membrane capacitance.** A higher capacitance smooths out voltage fluctuations and slows down changes in membrane potential.
- **EI, EK, ENa: Nernst potentials** for leak, potassium, and sodium ions. These are the driving forces for their respective currents.
- **g_na, g_kd: Maximal conductances** for fast Na⁺ and delayed-rectifier K⁺ channels. They are fundamental for generating action potentials.
- **gl: Maximal leak conductance.** It sets the resting membrane potential.
- **g_AHP:** Maximal conductance of the **after-hyperpolarization (AHP) current**. This current contributes to spike-frequency adaptation, where the firing rate decreases during a sustained stimulus.
- **VT:** Firing threshold offset. A more positive value makes the neuron less excitable.
- **sigma:** Standard deviation of the noise term. It introduces stochasticity, mimicking random synaptic inputs or channel fluctuations.
- **Tau_max:** Decay factor of the AHP current.
- **I_inj:** Injected current. It's a non-stochastic way to excite the neuron.
- **we_AMPA, we_NMDA:** Relative contribution of AMPA and NMDA channels to synaptic weight.
- **g_ampa, g_nmda:** Synaptic conductances for AMPA and NMDA receptors.
- **E_ampa, E_nmda:** Reversal potentials for AMPA and NMDA currents.

Synapse Parameters (get_Synparam)

These parameters regulate the strength and dynamics of synaptic transmission, including short-term plasticity and modulation by gliotransmitters.

- **E_lper**: Percentage of excitatory connections.
- **Omega_d: Depression rate**. It determines how quickly the pool of available neurotransmitter vesicles recovers after a spike.
- **Omega_f: Facilitation rate**. It determines how quickly the release probability recovers.
- **U_0_star: Basal synaptic release probability**. This is the initial probability that a presynaptic spike will cause a neurotransmitter release.
- **alpha**: Gliotransmitter modulation factor. This parameter determines the extent to which astrocytic gliotransmitter release affects the synaptic release probability, linking the astrocyte and synapse dynamics.
- **Omega_c**: Neurotransmitter clearance rate from the synaptic cleft.
- **rho**: Volume ratio of the synaptic cleft to the vesicle.
- **Y_T**: Total amount of neurotransmitter in a vesicle.
- **tau_rise_NT, tau_decay_NT**: Time constants for neurotransmitter concentration changes in the cleft.
- **O_G, Omega_G**: Binding and unbinding rates of gliotransmitters to presynaptic receptors.
- **tau_ampa, taus_nmda, taux_nmda**: Time constants for AMPA and NMDA receptor kinetics.
- **alpha_nmda**: NMDA channel activation rate.
- **tau_d**: Time constant for synaptic depression.
- **U**: Synaptic utilization (release probability).
- **tau_f**: Time constant for synaptic facilitation.
- **alpha_ampa_kin, beta_ampa_kin**: Rate constants for AMPA receptor binding and unbinding.
- **alpha_nmda_kin, beta_nmda_kin**: Rate constants for NMDA receptor binding and unbinding.
- **Xi**: Synaptic efficacy.
- **conn_prob**: Connection probability in a random network.
- **slope, intercept**: Parameters for a distance-dependent connection scheme.